

the membrane alters the membrane potential. The change in membrane potential is called a **receptor potential**, and conversion of the stimulus to a receptor potential is known as **sensory transduction**. Note that receptor potentials are graded potentials: Their magnitude varies with the strength of the stimulus.

Transmission

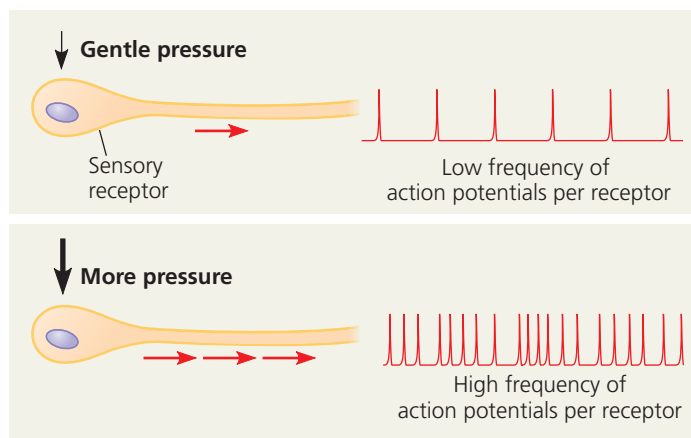
Sensory information travels through the nervous system as action potentials. A sensory receptor that is also a neuron generates action potentials that travel along an axon extending into the CNS (see Figure 50.3). In contrast, a non-neuronal sensory receptor does not itself generate action potentials, but instead conveys information to an afferent neuron via a chemical synapse. Because this chemical signaling alters the rate at which the afferent neurons produce action potentials, the sensory information generated also enters the CNS in the form of action potentials.

The size of a receptor potential increases with the intensity of the stimulus. If the receptor is a sensory neuron, a larger receptor potential results in more frequent action potentials (**Figure 50.4**). If the receptor is not a sensory neuron, a larger receptor potential usually causes the receptor to release more neurotransmitter.

Many sensory neurons spontaneously generate action potentials at a low rate. In these neurons, a stimulus does not switch the production of action potentials on or off, but instead changes *how often* an action potential is produced, alerting the nervous system to changes in stimulus intensity.

Processing of sensory information can occur before, during, and after transmission of action potentials to the CNS. In many cases, the *integration* of sensory information begins as soon as the information is received. Receptor potentials produced by stimuli delivered to different parts of a sensory receptor cell are integrated through summation, as are

▼ **Figure 50.4** Coding of stimulus intensity by a single sensory receptor.



postsynaptic potentials in neurons that receive input from multiple sensory receptors. As you'll see shortly, sensory structures such as eyes also provide higher levels of integration, and the brain further processes all incoming signals.

Perception

When action potentials reach the brain via afferent neurons, circuits of neurons process this input, generating the **perception** of the stimulus. An action potential triggered by light striking the eye has the same properties as an action potential triggered by air vibrating in the ear. How, then, do we distinguish sights, sounds, and other stimuli? The answer lies in the connections that link sensory receptors to the brain. Action potentials from sensory receptors travel along neurons that are dedicated to a particular stimulus; these dedicated neurons form synapses with particular neurons in the brain or spinal cord. As a result, the brain distinguishes stimuli such as sight and sound solely by the path along which the action potentials have arrived.

Perceptions—such as colors, smells, sounds, and tastes—are constructions formed in the brain and do not exist outside it. So if a tree falls and no animal is present to hear it, is there a sound? The falling tree certainly produces pressure waves in the air, but if sound is defined as a perception, then there is no sound unless an animal senses the waves and its brain perceives them.

Amplification and Adaptation

The transduction of stimuli by sensory receptors is subject to two types of modification—amplification and adaptation. **Amplification** refers to the strengthening of a sensory signal during transduction. The effect of amplification can be considerable. For example, an action potential conducted from the eye to the human brain has about 100,000 times as much energy as the few photons of light that triggered it.

Amplification that occurs in sensory receptor cells often requires signal transduction that involves enzyme-catalyzed reactions (see Concept 11.3). Because a single enzyme molecule catalyzes the formation of many product molecules, these pathways amplify signal strength considerably. Amplification may also take place in accessory structures of a sense organ. For example, the lever system formed by three small bones in the ear enhances the pressure associated with sound waves more than 20-fold before the stimulus reaches receptors in the innermost part of the ear.

Upon continued stimulation, many receptors undergo a decrease in responsiveness known as **sensory adaptation** (not to be confused with the evolutionary term *adaptation*). Sensory adaptation has very important roles in our perception of ourselves and our surroundings. Without it, you